



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC4231-1 – MEMS AND IC INTEGRATION

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the primary function of a micro actuator?**
A) To store data in a digital format B) To convert electrical signals into physical movement or force
C) To detect changes in the environment D) To transmit signals wirelessly
2. **What is one of the main advantages of integrated microsystems over traditional systems?**
A) Larger size B) Increased complexity
C) Miniaturization and reduced power consumption D) Higher manufacturing costs
3. **In a MEMS accelerometer, which part moves when acceleration occurs?**
A) The spring B) The diaphragm
C) The mass D) The resistive material
4. **In MEMS pressure sensors, what is typically used as the sensing element?**
A) Diaphragm B) Anchor
C) Conductive trace D) Magnet
5. **Which temperature range is used for the denaturation step in PCR?**
A) 35–40°C B) 50–55°C
C) 90–95°C D) 100–105°C
6. **Which of the following is a major advantage of fiber-optic sensors?**
A) Susceptibility to electromagnetic interference B) High power consumption
C) Immunity to electromagnetic interference D) Large size
7. **In MICROMIRROR ARRAY, Gold is used for**
A) Better mirror-reflectivity B) Electrodes
C) Sacrificial layer D) Electrical insulation
8. **Which of the following is a key advantage of MEMS magnetic micro relays?**
A) High power consumption B) Fast switching time and low power requirement
C) High voltage tolerance D) Large physical size
9. **Pre-deposition and drive-in are two steps of**
A) Diffusion B) Epitaxy
C) Metallization D) Ion implantation

10. In an integrated circuit, the SiO_2 layer provides
 A) Electrical connection to an external circuit B) Isolation
 C) Physical strength D) Conducting path
11. The main advantage of dry etching over wet etching is:
 A) It requires less time B) It is more selective and precise
 C) It uses fewer chemicals D) It is less expensive
12. What happens to the glass surface during anodic bonding?
 A) It becomes electrically conductive B) It melts at the bond interface
 C) It forms a thin oxide layer D) It becomes chemically inert
13. In an N-bit ADC, how many distinct values can the output signal take?
 A) N values B) 2^N values
 C) N^2 values D) 2^{N-1} values
14. In a differential amplifier, the output is ideally:
 A) Zero when both inputs are the same B) Double the input voltage
 C) Equal to the sum of the input voltages D) Proportional to the common-mode voltage
15. The lock range of a PLL refers to:
 A) The frequency range over which the PLL can maintain a stable lock B) The range of input signals the PLL can handle
 C) The bandwidth of the VCO D) The range of output frequencies the PLL can generate
16. Which of the following is not a characteristic of an instrumentation amplifier
 A) Low noise B) Low open-loop gain
 C) High CMRR D) High input impedances
17. Which of the following is an example of an inertial sensor?
 A) Accelerometer B) Thermistor
 C) Photodiode D) Barometer
18. What is the role of an air-gap in RF MEMS switches?
 A) To increase the capacitance B) To store electrical charge
 C) To enhance signal strength D) To allow for mechanical movement
19. Which of the following materials are commonly used in piezoelectric transducers?
 A) Lead zirconate titanate (PZT) B) Silicon
 C) Aluminum D) Copper
20. What is the term used for the phenomenon where a MEMS device gets stuck due to surface adhesion forces?
 A) Friction B) Stiction
 C) Torsion D) Creep

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the significance of miniaturization and microfabrication in designing smart systems.	4	L*2	1	1
	b) Draw the schematic of a bulk micro machined capacitive sensing accelerometer and explain its working principle.	4	L2	1	1
	c) Explain the working principle of a fiber-optic sensor with neat figures. Distinguish between Extrinsic sensor and intrinsic sensor.	8	L2	2	1
2.	a) Write short notes on Integrated Microsystems?	4	L2	1	1

EC4231-1

SEE - December 2025

- | | | | | | |
|-------|---|---|----|---|---|
| b) | Explain the structure and working of magnetic relay. | 4 | L3 | 2 | 1 |
| c) | Explain the structure and working of Conductometric Gas Sensors using neat diagrams and provide at least four examples of their use in modern applications. | 8 | L2 | 2 | 1 |
| 3. a) | What are smart materials and how are they used in smart structures? | 4 | L2 | 1 | 1 |
| b) | Discuss the working principle of an Electrostatic Comb-Drive. | 4 | L2 | 2 | 1 |
| c) | Explain the working principle of a piezoelectric inkjet actuator with neat schematics. | 8 | L2 | 2 | 1 |

Unit – II

- | | | | | | |
|-------|--|---|----|---|---|
| 4. a) | Draw and explain the basic setup of a sputtering system used in PVD. Compare it with thermal evaporation. | 8 | L2 | 3 | 1 |
| b) | Draw the circuit diagram of a difference amplifier using op-amp and prove that the output voltage of the difference amplifier is proportional to the difference between the two input signals. | 8 | L3 | 4 | 1 |
| 5. a) | What are photoresists? What are the important types of photoresists? Explain the effect of the type of photoresists used in lithography with neat process flow diagrams. | 8 | L2 | 3 | 1 |
| b) | Explain how a Wheatstone Bridge circuit can be used for precision measurement of unknown or varying resistance. Support your answer with a circuit diagram and relevant formulae. | 8 | L3 | 4 | 1 |
| 6. a) | Describe LIGA process steps with neat diagrams. List its advantages and disadvantages | 8 | L2 | 3 | 1 |
| b) | With the help of a neat block diagram explain the functioning of Phase-Locked Loops. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|--|----|----|---|---|
| 7. a) | Explain the principle of operation of a piezoresistive pressure sensor. Draw a labeled cross-section and write the governing equations. | 10 | L2 | 5 | 1 |
| b) | Define burst pressure of a pressure sensor? Calculate the burst pressure for the following cases for square silicon diaphragms with thickness h and diaphragm side length $2a$,
(i) $h=5\mu\text{m}$, $a=250\mu\text{m}$;
(ii) $h=10\mu\text{m}$, $a=250\mu\text{m}$.
Comment on the effect of diaphragm thickness on the burst pressure. Justify your answer. | 6 | L3 | 5 | 1 |
| 8. a) | Explain the structure and working of Microfabricated RF switch. | 8 | L2 | 5 | 1 |
| b) | With relevant sketches, explain the fabrication and working of a SOI based capacitive sensing accelerometer. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
